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Task Description:

Following exposure to spaceflight, upright posture can result in the inability to maintain adequate arterial pressure and cerebral perfusion. This may result in presyncope (lightheadedness) or syncope (fainting)during re-entry or egress. A significnat number of astronauts experience postflight orthostatic hypotension and presyncope during upright posture. The severity of this problem increases with increasing length of exposure to microgravity. About 20% of short-duration and 83% of long-duration crew members have this problem. It is expected that orthostatic hypotension will be severe on flights associated with exploration class missions. This project will measure the efficacy of midodrine in reducing the incidence and/ore severity of orthostatic hypotension in returning astronauts.x The purpose of this Supplemental Medical Objective (SMO) is to determine if long-duration spaceflight causes changes in cardiac repolarization processes that predispose to the development of ventricular dysrhythmias. If we find that there is an increased likelihood of dysrhythmias during long-duration spaceflight, this must be recognized and taken into account when planning mission activities. There is much anecdotal evidence concerning this issue, but little hard data. This SMO will collect data that will be used to assess this issue systematically. The possibility of ventricular dysrhythmias during long-duration spaceflight is the #1 priority on the NASA Cardiovascular Critical Path for Life Sciences.
The specific aims are 1) to determine if long-duration spaceflight affects cardiac repolarization processes; 2) if there are changes, to define possible mechanisms; and 3) to determine if the changes increase the likelihood of ventricular dysrhythmias.

While there are no definitive data showing that long-duration spaceflight generates serious ventricular dysrhythmias, there is much anecdotal data, which has been documented over many years, that is suggestive of cardiac electrical changes during these long flights. In the American experience during Skylab, all crewmembers exhibited some form of rhythm disturbance. Most were single premature ventricular contractions, but one crewmember experienced a five-beat run of ventricular tachycardia during a lower-body negative pressure protocol and another had periods of “wandering supraventricular pacemaker” during rest and following exercise. In addition, it was shown that the corrected Q-T interval (QTc), a marker of ventricular repolarization, was prolonged. The number of dysrhythmias was greater inflight than that shown by these men during their extensive preflight testing, and was higher than would be expected in a healthy population studied at random in Air Force studies (1,2).

Until the Mir missions, Skylab was the last time Americans had spent such extended time in space, but the risk of dysrhythmias remained a lingering concern. Because of this, inflight Holter monitoring was undertaken in the Shuttle era. In two reports, there were virtually no inflight dysrhythmias documented during either intravehicular or extravehicular operations in flights of 4 – 16 days (3,4).

As soon as Americans returned to long stays in space aboard the Mir space station, ventricular dysrhythmias again appeared. Once again PVC’s began to appear that had not been present before flight (5). A 14-beat run of ventricular tachycardia was documented (6); and the lay press reported several incidents of cardiac disturbances (6). In one incident, an American astronaut was sent to perform an EVA on the Mir because the cosmonaut who trained for the task was having ECG abnormalities.

More recently, we have found that after long-duration missions, Q-Tc intervals are prolonged in crewmembers that did not have prolonged Q-Tc intervals after their short flights. The limited inflight data available suggests that this occurs after about 21 days inflight (7). In the ground-based clinical population, prolongation of the Q-Tc interval is associated with an increased risk of ventricular arrhythmias in both cardiac patients (8,9) and in large population studies of healthy individuals (10,11). It is not known if Q-Tc interval prolongation is associated with an increased risk of ventricular dysrhythmias in astronauts.

All of these findings raise the concern that cardiac rhythm disturbances may become an issue during the long inflight tours of duty planned for space station and interplanetary missions. The degree to which spaceflight and its many variables can be considered dysrhythmogenic is not clear, but the possibility that serious cardiac rhythm disturbances might occur during spaceflight is a concern to the Cardiovascular IPT.

The QT interval is a measure of ventricular depolarization (QRS) and repolarization (T wave). The QRS complex is usually fixed in healthy individuals and does not change during long-duration spaceflight. Thus, changes in QT duration represent alterations in ventricular repolarization. The QT interval of the surface ECG is a spatial and temporal representation of all cardiac cellular action potentials. Not all cells within the heart share identical action potentials, therefore, a certain degree of variability or, inhomogeneity, in their repolarization time exists. The degree of inhomogeneity of repolarization directly correlates with the overall morphology of the QT interval (primarily the T wave) and in most cases with an increase in QT interval duration. A clear association between the magnitude of inhomogeneity of repolarization and the risk for the development of ventricular arrhythmias has been established (12,13).

Any factor that prolongs the action potential contributes to electrical instability during ventricular repolarization. When this happens, the membrane potential may abruptly reverse to depolarize and produce an early after depolarization (EAD). This may trigger a second action potential. If the second action potential excites the rest of the ventricle, a premature ventricular complex (PVC) will appear on the ECG. This pathophysiologic process may be repetitive and self-sustaining, thereby giving rise to a series of EADs that result in a ventricular arrhythmia.

Some conditions that can prolong the QTc interval are ischemic heart disease, autonomic dysfunction, bradycardia, electrolyte abnormalities, cardiac remodeling and medications that interfere with the cardiac potassium ion channels (14-18). Which of these factors are seen in long-duration astronauts? First, we know that astronauts develop autonomic dysfunction (19, 20, 21, 22). Second, we know that long-duration astronauts have a relative bradycardia compared to short-duration (22). Third, we have evidence of cardiac remodeling after long-duration flight (23). Fourth, there are medications on board that prolong Q-Tc interval, including ciprofloxacin, haldol, inderal, verapamil, zithromax, Zoloft and nortriptyline. It is possible that this combination of factors may create an environment in which QT intervals will prolong.

Prolongation of Q-Tc interval does not itself guarantee an increase in ventricular dysrhythmias. For example, sleep, hypothyroidism, and use of the antiarrhythmic drug amiodarone all prolong Q-Tc without increasing the incidence of ventricular dysrhythmias. It is possible that spaceflight presents a similar situation. However, at this time, that determination cannot be made, due to lack of data. Therefore, the data must be collected. We propose to study cardiac electrical as well as functional changes in response to long-duration spaceflight. No single measurement will be sufficient to address this complex situation. We plan to perform 48-hour Holter monitoring, 12-lead ECG’s, blood analyses, complete diagnostic echocardiograms, plasma and red blood cell volumes, hemodynamic

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will help to understand the mechanisms of orthostatic hypotension in both astronauts and patients. x
Task Progress:	The postflight portion of the study has been completed. We received JSC CPHS approval to procedd with phase II of the investigation to give midodrine at TIG. We have pitched the investigation to 2 crews, but have no volunteers yet.
Bibliography Type:	Description: (Last Updated: 04/15/2009)
Articles in Peer-reviewed Journals	Platts SH, Ziegler MG, Waters WW, Mitchell BM, Meck JV. "Midodrine prescribed to improve recurrent post-spaceflight orthostatic hypotension." Aviat Space Environ Med. 2004 Jun;75(6):554-6. PubMed PMID: 15198285 , Jun-2004